



Elementary Science

Uses of Electricity

Level 1

Topic 10: Electricity & Magnets | Worksheet 1

Name: _____

Date: _____

A Match each device to its job.



lamp

• •

makes moving air



fan

• •

shows pictures and sound



toaster

• •

makes light



television

• •

heats bread

B Which things use electricity?



phone



book



television



bicycle



fan



lamp



C Safe or unsafe?

1. Keep drinks away from sockets.

Safe

Unsafe

2. Put fingers into a socket.

Safe

Unsafe

3. Tell an adult about a damaged wire.

Safe

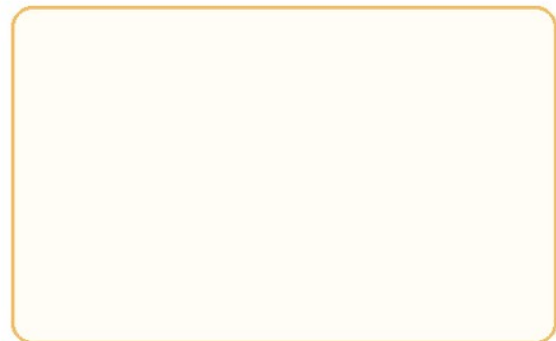
Unsafe

4. Use electrical devices with wet hands.

Safe

Unsafe

D Challenge: Draw an electrical c



What does it do?

Electricity helps many things work!



Elementary Science

Simple Circuits

Level 2

Topic 10: Electricity & Magnets | Worksheet 2

Name: _____

Date: _____

A Match each circuit part to its job



battery

turns the circuit on or off



wire

changes electrical energy into light



bulb

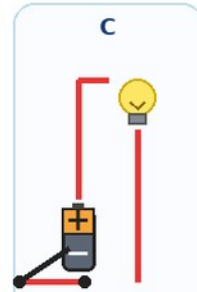
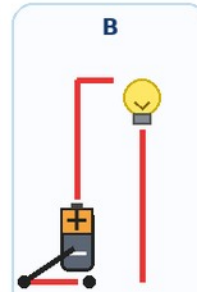
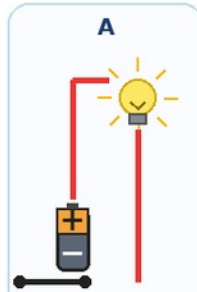
provides electrical energy



switch

carries current around the circuit

B Which circuit lights the bulb?



A complete, closed circuit lets current flow around the whole loop.

C Complete the sentences.

Word bank: battery wire bulb switch complete

1. The _____ provides electrical energy.

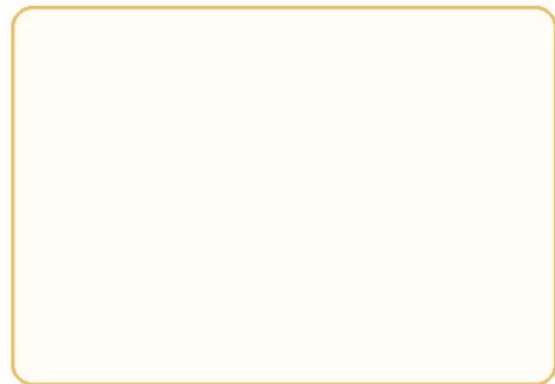
2. A _____ carries current.

3. A _____ can open or close a circuit.

4. The _____ changes electrical energy into light.

5. The bulb lights only when the circuit is _____.

D Challenge: Draw a complete circuit



Use a battery, wires, a bulb and a closed switch.

A complete circuit makes a continuous path!



Elementary Science

Conductors & Insulators

Level 3

Topic 10: Electricity & Magnets | Worksheet 3

Name: _____

Date: _____

A Look at the test results.

Material	Bulb lights?
metal paper clip	YES
coin	YES
plastic spoon	NO
wood block	NO
rubber	NO

If the bulb lights, the material allows electricity to pass through.

B Match each word to its meaning

conductor



often used to cover wires

insulator



lets electricity pass through easily

metal



often a good conductor

plastic



does not let electricity pass through easily

C Use the results to answer.

1. Which two tested materials were conductors?

2. Which tested materials were insulators?

3. Which material type is often used inside wires?

4. Which material is useful around the outside of wires?

D Explain: Why do wires have two



metal core

plastic coating

Conductors let current through; insulators help keep us safe.



Elementary Science

Magnets & Electromagnets

Level 4

Topic 10: Electricity & Magnets | Worksheet 4

Name: _____

Date: _____

A Magnetic or non-magnetic?

-  paper clip ☐ magnetic ☐ not magnetic
-  iron nail ☐ magnetic ☐ not magnetic
-  coin ☐ magnetic ☐ not magnetic
-  wood block ☐ magnetic ☐ not magnetic
-  plastic spoon ☐ magnetic ☐ not magnetic
-  steel screw ☐ magnetic ☐ not magnetic

B Poles: attract or repel?

- ☐ N ☐ S ☐ attract ☐ repel
- ☐ N ☐ N ☐ attract ☐ repel
- ☐ S ☐ S ☐ attract ☐ repel
- ☐ S ☐ N ☐ attract ☐ repel

Unlike poles attract. Like poles repel.

C Electromagnet investigation.

Same iron nail and wire; only the number of cells changed



Cells	Paper clips picked up
1 cell	3
2 cells	6

3 cells
What pattern do you notice?

D Answer the questions.

- What is wrapped around the iron nail?

- What provides electrical energy?

- Which setup was strongest?

- What happens when the current is switched off?

Magnets can push or pull — electromagnets use electricity!



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3. Tell an adult about a damaged wire.

Safe

Unsafe

4. Use electrical devices with wet hands.

Safe

Unsafe

D Challenge: Draw an electrical c



Example: television

What does it do?

It shows pictures and sound.

Electricity helps many things work!



Elementary Science

Simple Circuits

Level 2

Topic 10: Electricity & Magnets | Answer Sheet 2

Name: _____

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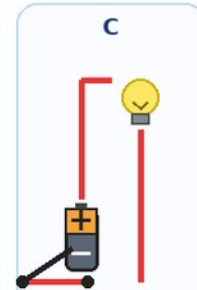
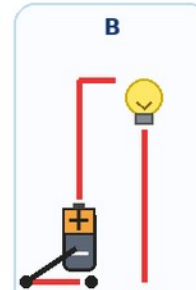
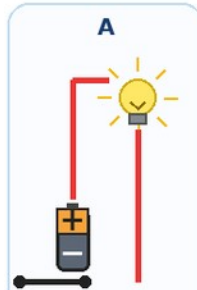
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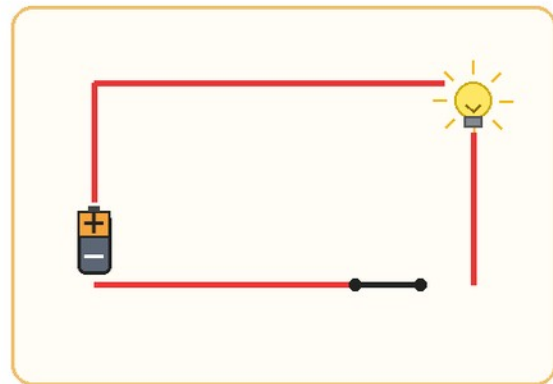
2. A _____ carries current.
wire

3. A _____ can open or close a circuit.
switch

4. The _____ changes electrical energy into light.
bulb

5. The bulb lights only when the circuit is _____.
complete

D Challenge: Draw a complete circuit



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A complete circuit makes a continuous path!



Elementary Science

Conductors & Insulators

Level 3

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insulator

lets electricity pass through easily

metal

often a good conductor

plastic

does not let electricity pass through easily

C Use the results to answer.

1. Which two tested materials were conductors?
paper clip and coin

2. Which tested materials were insulators?
plastic, wood and rubber

3. Which material type is often used inside wires?
metal

4. Which material is useful around the outside of wires?
plastic or rubber

D Explain: Why do wires have two



metal core

plastic coating

The metal core conducts electricity. The plastic coating is an insulator that helps keep the current away from our hands.

Conductors let current through; insulators help keep us safe.



Elementary Science

Magnets & Electromagnets

Level 4

Topic 10: Electricity & Magnets | Answer Sheet 4

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Date: _____

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-  plastic spoon magnetic not magnetic
-  steel screw magnetic not magnetic

B Poles: attract or repel?

- N S attract repel
- N N attract repel
- S S attract repel
- S N attract repel

Unlike poles attract. Like poles repel.

C Electromagnet investigation.

Same iron nail and wire; only the number of cells changed



Cells	Paper clips picked up
1 cell	3
2 cells	6

3 cells

9

What pattern do you notice?

More cells made the electromagnet stronger.

D Answer the questions.

- What is wrapped around the iron nail?
wire
- What provides electrical energy?
the cells / battery
- Which setup was strongest?
3 cells
- What happens when the current is switched off?
the electromagnet stops attracting

Magnets can push or pull — electromagnets use electricity!